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RESEARCH ARTICLE

Prediction of Pilot Performance During Startle Events Based on Neuropsychophysiological Features of Stress Resilience and Cognitive Task Scores

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ABSTRACT The disruptive impact of the startle effect on pilots can lead to potentially fatal outcomes, highlighting the importance of identifying novel predictors of pilot performance during unexpected and unpredictable startle events. Accordingly, this study investigates the potential of optimally selected stress resilience features and cognitive task scores to predict pilot performance during startle events. Seventeen military pilot cadets participated in a two-phase experiment conducted over two weeks. In Phase 1, context-free stimulation paradigms including neuropsychophysiological measurements were administered. In Phase 2, specific flight simulator sessions with startle events were undertaken. Optimal neuropsychophysiological stress resilience features and cognitive task scores from Phase 1 were selected as predictors of pilot simulator performance in Phase 2 using multiple regression analysis (MRA) within a leave-one-out cross-validation (LOOCV) procedure. The obtained MRA models demonstrated significant predictive validity across left-out participants, as indicated by a Pearson correlation coefficient of 0.7039 ($p = 0.0034$) between the actual and predicted simulator performance, with a mean absolute error (MAE) of 2.5457° versus a dummy median regressor MAE of 3.4881°. The optimally predictive features/scores from Phase 1 were related to startle reflex habituation, multitasking ability, heart rate recovery after threat, and resting respiratory sinus arrhythmia. These findings may improve the accuracy of predicting pilot performance during startle events, with implications for pilot selection and training.

INDEX TERMS Cognitive task scores, feature engineering, flight simulation, human factors, physiological features, pilot performance prediction, startle effect, stress resilience.

I. INTRODUCTION

In aviation, startle effect is defined as a physiological, cognitive and behavioural response that is triggered by a sudden, intense, or threatening stimulus [1], [2], [3]. In the cockpit, the disruptive impact of the startle effect can be significant, particularly when it occurs during crucial flight tasks. This disturbance can endure from 0.1 to 3 seconds for simpler tasks and up to 10 seconds for more intricate motor activities [4]. Beyond its immediate distraction, startle events can elevate overall stress levels, leading to detrimental impact on cognitive functions like narrowed attention, diminished search behaviour, delayed reactions to peripheral cues, decreased vigilance, compromised problem-solving abilities, inflexible

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performance, and impaired working memory function [5]. The higher the intensity of startle events, the stronger the fight/flight reaction becomes, making it increasingly challenging to return to a deliberate analysis of the situation [6].

Considering the dangerous effects startle events can have in a cockpit [7], it is critical to have more reliable predictors of pilots' performance [8] during startle events. Many studies have investigated the relationship between different multimodal physiological features and simultaneously measured pilot performance metrics on flight simulators, particularly in the context of mental workload [9], [10], [11], [12], and other relevant pilot states [5], [13], [14]. However, research specifically investigating physiological indicators of stress resilience or vulnerability as predictors of job-relevant performance under stress in pilots and air traffic controllers remains limited, with only a few studies addressing these topics [15], [16], [17], [18].

To address this gap and better understand neurophysiology of startle effect in aviation, which can be elicited by different sensory modalities (e.g., auditory sounds, objects suddenly entering the visual field), it may be helpful to review the extensive independent body of literature regarding the basic acoustic startle reflex (ASR). Koch [19] defines ASR as an involuntary activation of motor pathways generated in the brainstem, primarily within the pontomedullary reticular formation, that is typically triggered by a brief, intense acoustic stimulus [20], [21]. The magnitude of ASR and its modulations, such as prepulse inhibition (PPI) and fear-potentiated startle (FPS), are well-known indicators of vulnerability to stress [15], [22], [23], [24], [25], [26], [27], [28], which is hypothesized to detrimentally impact pilot's job performance in highly stressful situations [15], [29], [30], [31] like unexpected and unpredictable startle events. This study aims to explore pilots' individual variability in ASR and related physiological stress resilience features as potential predictors of performance during startle events.

Additionally, a broad spectrum of cognitive, emotional, and motor abilities is essential for successful execution of pilots' realistic, complex, and stressful tasks [32], [33], [34], [35], [36], [37], [38], [39]. Accordingly, research in this paper also includes scores on selected relevant laboratory cognitive tasks as potential predictors of pilot's in-flight performance during startle events.

In line with these objectives, this study introduces a novel approach for predicting pilot performance in a flight simulator under startle conditions, based on optimally selected neuropsychophysiological stress resilience features and cognitive task scores. Recognizing that the startle effect can significantly impact pilot functioning, this approach emphasizes the importance of understanding an individual's stress response as a predictor of pilot's ability to manage sudden and unexpected challenges in the cockpit. By focusing on the physiological context-free laboratory assessment of stress resilience indicators—which include features related to resting autonomic functioning, ASR, and psychobiological allostasis [15], [26]—this study aims to evaluate the predictive value of relatively stable traits [40] for performance in job-specific, operationally relevant tasks. Although these physiological features and cognitive task scores are assessed in a controlled, context-free laboratory setting, they are proposed to predict pilot performance on flight simulator in complex, dynamic, and stress-intensive situations, such as startle events [15], [41], [42].

To investigate this, a two-phase experimental design was implemented. The first phase involved context-free stimulation paradigms, including a stress resilience paradigm and different cognitive tasks. The second phase examined pilot performance under startle conditions in a flight simulator environment, designed to simulate real-life operational demands.

This study addresses the following research questions:

RQ1: How well do multimodal neuropsychophysiological stress resilience features related to the acoustic startle reflex, resting autonomic activity, and allostatic load, combined with cognitive task scores, predict pilot performance during startle events in a flight simulator-based use case?

RQ2: Which specific multimodal neuropsychophysiological stress resilience features and cognitive task scores are the most effective predictors of pilot simulator performance during startle events?

II. BACKGROUND

A. COGNITIVE, EMOTIONAL, AND MOTOR INTERACTIONS RELATED TO PILOT PERFORMANCE DURING STARTLE EVENTS

The neurocognitive requirements needed for highly effective pilots' on-the-job performance depends on their individual emotional regulation, stress resilience, situational awareness, adaptability, precise hand-eye coordination and decision-making capacity [12], [33], [34], [35], [36], [43], [44]. Key cognitive abilities for effective pilot performance during startle events are related to working memory, attentional control, situational awareness, and multitasking [37], [38], in addition to key motor abilities related to reaction time, fine motor precision, and coordinated control [45]. These cognitive, emotional, and motor capabilities for successful pilot performance under a variety of operational stressors, including startle events, can be evaluated by a diverse set of generic, context-free stimulation paradigms. An individual's capacity to handle stress and adapt to challenging stressful environments can be evaluated through objective physiological indicators of stress resilience related to resting autonomic functioning, startle reflex, and psychobiological allostasis [15], [26], [28]. The following list summarizes some of the relevant cognitive abilities which can be tested in laboratory settings by standardized paradigms:

- Working memory, along with its capacity and updating ability, can be assessed using the standard n-back test [46], [47], [48].
- Cognitive flexibility, expressed through processes of inhibition, attention, processing speed, and cognitive

control, can be assessed, for example, using the standard Stroop test [49], [50], [51].
- Multitasking abilities, which reflect the limits of human cognitive processing capacity, can be assessed using dual-task paradigms [52], [53], [54], [55].
- Information processing speed, working memory, attention, and concentration can be assessed using the standardized Paced Auditory Serial Addition Task (PASAT) [56].
- Oculomotor fatigue, visual-motor abilities, and attentional distribution can be assessed through various oculometric paradigms, such as multiple object tracking (MOT) tasks [57], [58], [59].

Accordingly, the prediction of pilot performance in a specific operationally relevant startle setting, represented in a flight simulator session, is hypothesized to be most accurate when relying on selected context-free stimulation paradigms that recruit similar underlying neural circuits and processes. Cognitive, emotional, and motor abilities that are targeted by generic context-free stimulation paradigms and complex simulator use cases from the articles reviewed in this chapter are listed in Table 1.

B. MULTIMODAL PHYSIOLOGICAL FEATURES IN ASSESSMENT OF PILOT PERFORMANCE

Majority of aviation studies linking multimodal physiological features to pilot performance have been conducted during simulated flight tasks. These studies typically involve correlational analyses of physiological responses recorded during simulation and corresponding performance metrics [5], [13], [61], [62], [63], [64]. For example, Kinney and O'Hare [5] reported an increased heart rate (HR) and pupil dilation in pilots facing unexpected engine failures, which were associated with impaired task performance. Similarly, a body of machine learning–driven studies aim to detect or predict startle and surprise responses in flight simulators using physiological signals [65. For example, Binias et al. [72] demonstrated that pre-event EEG activity could predict pilots' reaction times to unexpected events with a high temporal precision. Additional studies focused on the relationship between pilot performance and physiological features are related to pilot mental states other than startle effect, like mental workload, mental fatigue, and stress [9], [10], [11], [12], [35], [73], [74], [75], [76], [77].

However, few studies have examined whether physiological features assessed before a flight simulation, particularly in a context-free laboratory setting, can predict a pilot's subsequent task performance. Deniel et al. [78] conducted a two-phase study examining the susceptibility to stress-induced incapacitation via pre-assessments and flight simulator testing with startle induction. Their findings emphasized the individual variability in startle susceptibility. Causse et al. [16] found that prefrontal activation during a working memory task predicted subsequent flight performance, while Hidalgo-Muñ oz et al. [79] linked stable electrodermal activity and personality traits to greater stress resilience in the cockpit. Similarly, Miyatsu et al. [17] demonstrated that post-stress HR and heart rate variability (HRV) measures reflected executive function and stress tolerance in a military selection context, with elite candidates showing higher parasympathetic and lower sympathetic activation. Together, these findings underscore an importance of individual stress response patterns as valuable predictors of operational performance. Complementing these physiological measures, several studies have shown that cognitive test scores [34], [39], [80], [81], [82] and personality traits [83], [84] are related to flight performance. Notably, Mouratille et al. [18] found that baseline HRV predicted adaptability under stress, echoing the relevance of physiological resilience for performance prediction.

The current study advances this line of research by systematically examining whether physiological features of stress resilience and cognitive task performance scores, evaluated in a context-free laboratory environment, can predict pilot simulator performance in operationally relevant scenarios involving startle events. Unlike many previous studies that assess pilots' physiological responses during simulated flights, in this work, pilots' stress resilience physiological features are evaluated before flight simulation—specifically acoustic startle reflex, resting autonomic functioning, and allostasis—and used as predictors of their performance in a demanding startle-related flight simulator use case. This approach builds upon earlier work in the domain of air traffic control [15], which demonstrated the predictive value of stress resilience physiological features for simulator performance. It also aligns with [78] in recognizing the significance of individual differences in stress response and their impact on operational readiness. Altogether, by integrating multimodal, context-free physiological and cognitive assessments within a two-phase experimental design, this study contributes to the literature seeking to identify trait-based physiological and cognitive predictors of performance in high-stakes, operationally relevant aviation scenarios.

III. METHODS

A. STUDY PARTICIPANTS

Seventeen (17) male military pilot students (first and second year) from the University of Zagreb participated in this study (mean age = 19.88; SD = 0.99 years; age range = 19 – 22; mean flight experience = 12.5 hours, SD = 10 hours; range = 6 - 50). Due to technical issues during the data collection process, data from two participants were excluded from further analysis thus leaving 15 participants in the dataset to be analysed. All participants underwent a nationally standardized selection process for military pilot education and training, where they had to meet the psychological, health, physical, and security criteria for military pilot students. Also, as a part of nationally standardized selection process, they completed a flying evaluation lasting 5 to 7 weeks. Participation in the study was voluntary, and all participants signed an informed

TABLE 1. Overview of cognitive, emotional, and motor abilities targeted by the reviewed generic context-free stimulation paradigms and complex simulator use cases.

PARADIGM	ABILITIES TARGETED	LITERATURE
ASR eliciting paradigm	Stress, emotions, resilience	[15], [28], [59], [60]
N-back	Working memory: capacity & updating ability	[47], [48]
Oculometric paradigm	Fatigue, attentional capacity, visual tracking skills, working memory	[57], [58], [59]
PASAT	Information processing speed, working memory, attention, concentration	[56]
Dual Stroop	Attention, working memory, executive functioning, interference, inhibition, processing speed, cognitive control, multitasking abilities	[50], [51], [54], [55]
Complex simulator use cases	Multitasking abilities, situational awareness, cognitive flexibility, emotional regulation, focus, hand-eye coordination, executive functioning	[33]-[38]

consent according to the guidelines of the University of Zagreb. The study was approved by the Ethics Committee of the University of Zagreb Faculty of Transport and Traffic Sciences.

B. EXPERIMENTAL PROTOCOL

The experimental protocol consisted of two different phases. In **Phase 1**, the participants went through following context-free stimulation paradigms:

- **Standardized laboratory ASR eliciting paradigm** for the physiological assessment of stress resilience is modification of that from Ćosić et al. [28]. The paradigm started with a three-minute resting block to establish baseline measures, followed by a block where participants are exposed to 10 loud 40-ms 108-dB (sound pressure level, SPL) white-noise auditory startle (AS) stimuli. In the PPI block, a 75 dB acoustic prepulse was preceding AS stimulus by 120-ms lead time. This block consisted of 4 referent (pulse-alone) and 10 PPI startle stimuli. The FPS was elicited by a danger block, where participants were exposed to 4 unpleasant composite stimuli, in addition to AS stimuli while the cue ''DANGER'' was written on the screen. The composite stimulus was a combination of a 250-ms air blast and a loud 40-ms 108 dB (SPL) white-noise auditory stimulus. The paradigm concluded with a two-minute recovery block to monitor how quickly physiological responses return to baseline after the stress phase. The total duration of the paradigm was approximately 13 minutes.
- **Oculometric paradigm**, used in this research, was introduced by Ćosić et al. [59] and has seven alternating blocks of reflexive dot pursuit (RDP) task and multiple object tracking (MOT) task, starting with an RDP block. The total duration of the paradigm was 10 minutes.
- A visual variant of PASAT, **paced visual serial addition task** (PVSAT), was used in this research. Participants were presented with a series of single-digit numbers in rapid succession. Their task was to add each new number mentally to the one immediately preceding it and verbally report the sum. Participants were presented with 4 blocks of 40 single-digit numbers. In later blocks, single-digit numbers were displayed faster than in the beginning, thus increasing the difficulty of the task. The total duration of the paradigm was 7 minutes.
- **The Stroop task** used in this study was a colour-word interference task consisting of both congruent and incongruent trials. Visual stimuli were Croatian colour words displayed in font colours that were either congruent or incongruent with the word meaning. Participants were instructed to respond to the font colour while ignoring the word content. A total of 75 stimuli were presented, each shown for 0.5 s, followed by a white screen for 0.75 s. The total task duration was approximately 2 minutes.
- **Dual Stroop paradigm** consisted of the Stroop task and a concurrent task of monitoring and counting high-pitched auditory stimuli. Participants were required to perform 6 blocks of 15 congruent and incongruent Stroop tasks while being exposed to a series of low and high-pitched auditory stimuli. After each block of 15 Stroop tasks, the participants were required to say the number of high tones they counted in the current block. The total duration of the dual task paradigm was approximately 3 minutes.
- The design of the **n-back task** used in this research was the same as in Kesedžić et al. [48] and Gambiraža et al. [85], but with only two difficulty levels (n = 1 or 3) instead of the original four (n = 0, 1, 2, or 3). The stimuli presented were letters. The subjects underwent a total of 150 trials of the n-back task, evenly distributed across two difficulty levels and six task blocks. The total duration of the protocol was approximately 10 minutes. Additionally, the order of difficulty levels was randomized among participants.

At the end of each task, participants were given a break before proceeding to the next task to reduce cognitive fatigue. Additionally, during this phase of the experiment, standardized psychological questionnaires were administered, but their analysis is not in the scope of this paper. Several days later

(6.88 ± 2.71 days), participants completed a **Phase 2**, which took place on a flight simulator and consisted of:

- Familiarization with the flight simulator
- **Pitch tracking task** in which participants were required to achieve the commanded attitude by tracking the commanded pitch angle shown on the artificial horizon (Fig. 1). The candidates were told that the airspeed should be kept constant during the tasks and that only the deviation from the commanded pitch angle would be measured to assess their performance. Candidates did not have to keep track of the flight altitude because the commanded signal was defined in a way that following its average value kept the aircraft at a constant altitude. The total duration of the task was approximately 3 minutes.
- **Pitch tracking task with startle events** that consisted of the pitch tracking task coupled with three startling events occurring during the task (Fig. 2). At certain points in the simulation, a strong wind of 40 m/s, aligned with the aircraft's longitudinal axis, was triggered. Two gusts were modelled as tailwinds and one as a headwind. Each gust caused a rapid change in the airspeed and a slight 2° change in the pitch angle. The disturbance in airspeed and pitch angle increased gradually over the following two seconds, with the direction of the wind gust determining whether the change was positive or negative. Simultaneously, a loud thunder sound was played through the candidates' headphones. Immediately following each startling event, candidates were instructed, via audio cues delivered through headphones, to vocally report a specific flight parameter—either the altitude, flight speed or vertical speed, which served as a distraction from the main pitch tracking task. The total duration of the task was approximately 3 minutes.

The presented flight simulator use case, which comprised a pitch tracking task with and without startle events, was previously validated by a group of more experienced pilots [86]. While physiological signals were recorded during the second phase, their analysis is beyond the scope of the present study. Prior to the start of each phase, participants completed a survey assessing sleep quality, caffeine intake before measurement, and other variables that could potentially influence the measured multimodal neuro-psycho-physiological parameters. For the complete experimental design, see Fig. 3.

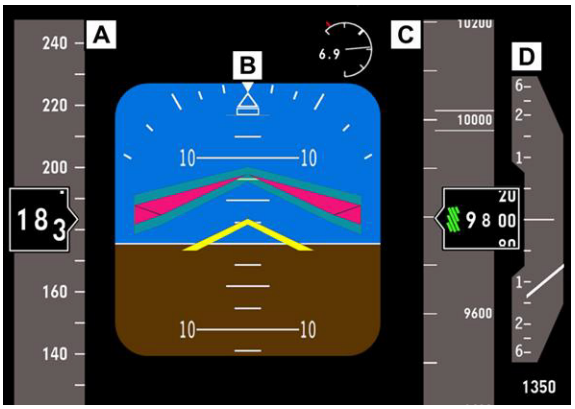


FIGURE 1. Flight instrument screen presented to the participants in both pitch tracking tasks. Letters denote a specific flight instrument: A – airspeed in knots, B – artificial horizon with pitch tracking indicators, C – altitude in feet, D – vertical speed in feet per minute.

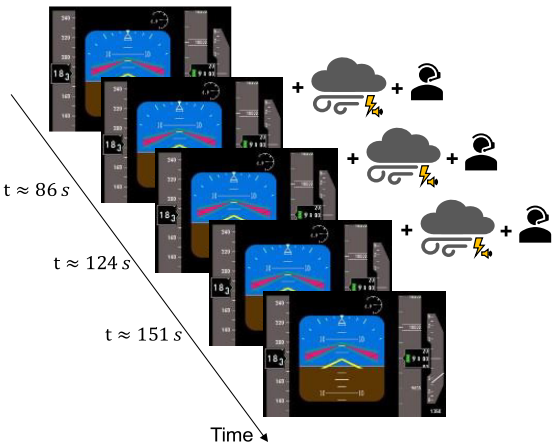


FIGURE 2. A schematic overview of pitch tracking task with startle events. Startle events, consisting of a strong wind gust and a loud thunder, were immediately followed by the candidate's vocal report of a specific flight parameter.

C. EXPERIMENTAL SETUP

1) PHASE 1: SETUP FOR GENERIC CONTEXT-FREE STIMULATION PARADIGMS

The setup for neuropsychophysiological measurements used during execution of context-free stimulation system with the accompanying modules for collecting the paradigms consisted of a BIOPAC MP150 data acquisition electrocardiography (ECG), electromyography (EMG) of orbicularis oculi, electrodermal activity (EDA), and respiratory (RESP) data, as a part of an integrated multimodal stimulation and measurement system [60] (Fig. 4). Physiological signals were visually inspected to verify that the sensors were operational, indicated by identifiable QRS complexes in ECG, skin conductance responses in EDA, inhalation–exhalation patterns in RESP, and blink-related amplitude increases in EMG. Visual inspection of physiological signals after electrode placement, lasting one minute, further helped the participants' signals to settle to a basal rest level before the start of the paradigm.

2) PHASE 2: FLIGHT SIMULATOR SETUP

A six-degrees-of-freedom flight model developed in MATLAB Simulink (MathWorks Inc.) was used in the experiment. Modelled aerodynamic coefficients were linearized for small angles of attack. The aircraft model used in the experiment was a generic fighter in canard configuration with retracted flaps and landing gear and without external fuel tanks or armament. Although the flight model can be extended to support supersonic flight, the modelled aircraft

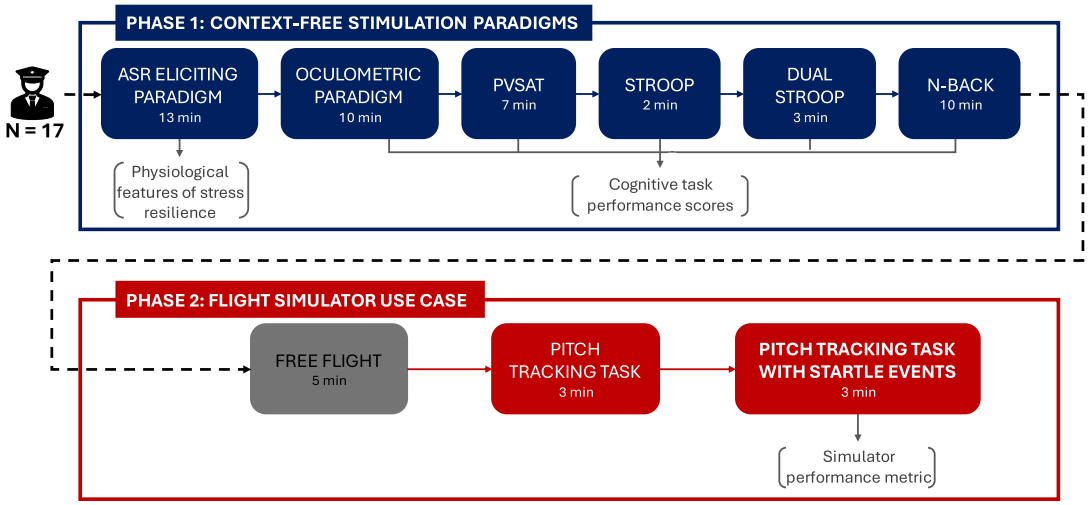


FIGURE 3. Schematic overview of the experimental protocol. The study consisted of two distinct phases separated by a break of several days (mean = 6.88 ± 2.71 days). Phase 1 involved context-free stimulation paradigms to assess neuropsychophysiological stress resilience and cognitive performance. Phase 2 consisted of a flight simulator use case designed to evaluate pilot performance under startle conditions.

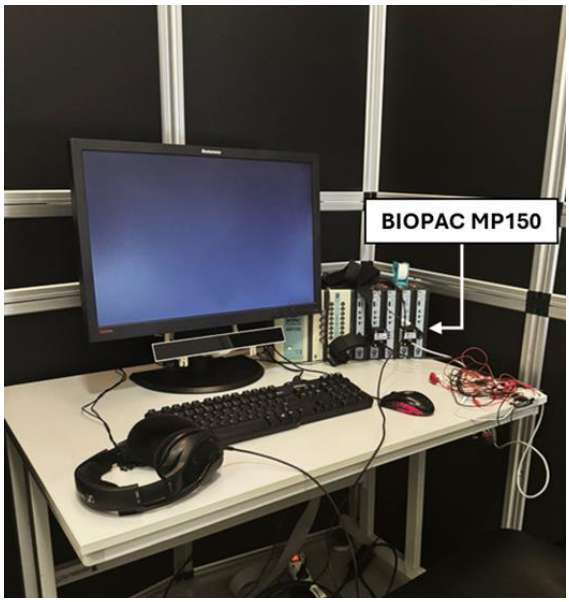


FIGURE 4. Setup used during Phase 1 for conducting generic context-free stimulation paradigms, featuring the BIOPAC MP150 data acquisition system (highlighted), equipped with modules for ECG, EMG, EDA, and respiration as part of a multimodal stimulation and recording system.

was limited to subsonic speeds during the experiment. This limitation was not an issue because the flight speeds in the experiment were low, starting at 90 m/s (175 knots), to enhance the influence of wind gusts on the aircraft's attitude and speed. In the modelled aircraft, the stick deflection directly controlled the angular velocity of the aircraft along a certain axis, while the wind speed function transitioned from zero to its maximum value over a period of one second. Simultaneously with the wind onset, the same signal defining the wind speed, but with a different amplification, was added to the flight stick deflection signal, which resulted in a mild change in the pitch angle at the time of the wind onset. The visual system of the flight simulator consisted of three XGA-resolution projectors and a cylindrical screen with a 2.5 m radius in front of the screen. The visual field covered a horizontal angle of 180° and a vertical angle of 50°. The seat, stick, throttle, rudder pedals and a monitor used for displaying flight instruments were located on a fixed-base in front of the screen (Fig. 5).

D. DATA PROCESSING AND FEATURE EXTRACTION

In the further analysis, the study focused on the physiological signals recorded during the ASR eliciting paradigm, including ECG, EMG, EDA, and respiration measurements. Additionally, scores were obtained from cognitive tasks, specifically from the MOT, PVSAT, Stroop task, dual Stroop task, and the N-back task. In the context of the flight simulator, the analysis centred on metrics derived from the simulation performance only.

Physiological data processing followed the same approach as described in Šarlija et al. [15]. Stress resilience features were divided into resting, startle reflex and allostatic features in accordance with prior research on stress resilience which categorized physiological responses into distinct domains to capture different aspects of stress reaction, adaptation, and recovery [15], [26], [28]. A complete list of stress resilience features is presented in Table 2, while complete cognitive task scores and simulator performance metric are presented in Table 3. For detailed statistical analysis of the stress resilience features see [15] and [28]. All features were normalized using z-score normalization prior to any analyses to ensure consistency in scale and comparability across variables.

E. STATISTICAL, CORRELATION AND REGRESSION ANALYSIS

An initial exploratory analysis was conducted on the full dataset using Pearson and Spearman correlation coefficients



FIGURE 5. Participant engaged in a pitch tracking task in a flight simulator setup with a cylindrical display and flight instrumentation interface.

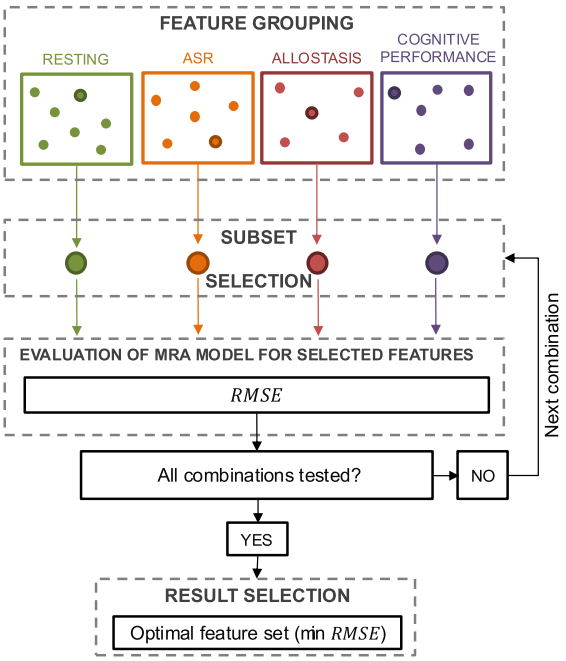


FIGURE 6. Schematic representation of the optimization algorithm. Subsets of features are selected and evaluated using MRA models, with performance assessed by RMSE.

to examine the relationships between individual stress resilience physiological features and cognitive task scores from Phase 1 with simulator performance from Phase 2. These correlations were visualized using scatter plots to illustrate the patterns and relationships specific to our dataset. This exploratory analysis was performed independently of the modelling process and was not used to guide feature selection. All data processing and analysis were conducted in Matlab R2024b.

Subsequently, a leave-one-out cross-validation (LOOCV) framework was implemented to find optimal stress resilience physiological features and cognitive task scores as predictors of simulator performance. In each iteration of the LOOCV, while one participant was left out, correlation-based feature preselection was conducted within each of the four distinct feature groups: resting physiological features, startle reflex physiological features, allostatic physiological features, and cognitive task scores. The top five features having the highest absolute correlations with the simulator performance were identified within each group. An optimization algorithm was applied to select the best combination of four predictors—one from each group—that minimized the root mean squared error (RMSE) of the resulting multiple regression model (Fig. 6).

The selected model in each iteration was then used to predict the simulator performance of the left-out participant, and the absolute difference between the predicted and actual performance was recorded. These errors were aggregated across LOOCV iterations to calculate the mean absolute error (MAE) as a metric for the evaluation of the predictive accuracy, and a Pearson correlation coefficient between predicted and actual performance values was calculated as well. This approach ensured the generalizability and representation of predictors from all four theoretically motivated feature groups, while limiting overfitting. These constraints also helped mitigate common challenges associated with MRA, including multicollinearity among predictors—which is known to occur among physiological features of stress resilience [15]—and the difficulty of selecting relevant predictors from a large pool of potentially useful candidates [88]. An overview of the full LOOCV data analysis workflow is presented in Fig. 7.

IV. RESULTS

A. STATISTICAL AND CORRELATION ANALYSIS RESULTS

Pearson and Spearman correlation results for the selected stress resilience features and cognitive task scores from Phase 1 with simulator performance under startle conditions from Phase 2 are presented in Table 4. The table includes the top three features per each of the four groups, along with their original value ranges and correlation coefficients (r) and significance levels (p) for both Pearson and Spearman correlations. This exploratory correlation analysis served as a preliminary step to characterize the relationships within the dataset. Among these features, the strongest correlation was observed for dStroopTones ($r_{Pearson} = -0.5666$, $r_{Spearman} = -0.5085$). Additionally, simulator performance, defined as the overall error in pitch tracking, demonstrated considerable between-subject variability, ranging from 2.33 to 14.36 degrees (mean overall error = 6.91; SD = 4.13), reflecting differing levels of tracking ability under startle conditions across participants. Fig. 8 displays scatter plots

TABLE 2. An overview of physiological features of stress resilience calculated from the standardized laboratory ASR-eliciting paradigm conducted in Phase 1.

GROUP OF FEATURES		FEATURE	FEATURE DEFINITION	LITERATURE
Resting Features	Resting cardiac activity	IBI	Inter-beat interval	[15], [28], [48], [59]
		HR	Heart Rate	
		RMSSD	The root mean square of successive differences between heartbeats	
		RMSnSD	The root mean square of normalised successive differences between heartbeats	
		RSA_1	Mean peak-to-trough difference in IBI times determined on a breath-by-breath basis	
		RSA_2	Correlation between uniformly resampled (10Hz) baseline-corrected IBI time series and respiratory effort signal in the time domain (Spearman's correlation)	
		RSA_3	Relative HRV power spectral density (PSD) estimate around the breathing frequency (±0.015Hz).	
		RSA_4	Correlation between the uniformly resampled HRV PSD estimate and respiratory PSD estimate (Spearman's correlation)	
		RSA_{PCA}	Linear combination of the 4 calculated RSA features that corresponds to the first PCA axis (71.43% of total variance explained)	
	Resting respiratory activity	BR	Breathing rate during rest	
		BA	Breathing amplitude during rest	
Acoustic Startle Reflex (ASR) Features	Startle reactivity	SR_{emg}	EMG-based ASR magnitude for the first 3 startle probes divided by the mean EMG response magnitude in activation of spontaneous blinks during the *Resting block*	
		SR_{eda}	EDA-based ASR magnitude for the first 3 startle probes divided by the mean tonic EDA value during the *Resting block*	
		SH_{emg}	Slope of the linear model fitted to normalised EMG responses to first 10 startle probes	
		SH_{eda}	The rate of the exponential model fitted to normalised EDA responses to first 10 startle probes	
	Startle modulation	PPI_{emg}	Mean EMG ASR magnitude for 10 PPI startle probes, divided by the mean ASR magnitude for 8 *Reference* startle probes (last 4 startle stimuli from the habituation block, and 4 noise-alone startle stimuli from the PPI block)	
		PPI_{eda}	Calculated in the same way as PPI_{EMG}, but using the EDA-based ASR measures	
		FPS_{emg}	EMG-based measure of FPS is calculated as the mean EMG AS response magnitude for 10 FPS probes (delivered during the danger block), divided by the mean ASR magnitude for 8 *Reference* startle probes	
		FPS_{eda}	Calculated in the same way as FPS_{EMG}, but using the EDA-based ASR measures	
Allostatic Features	Reaction	IBI_{react}, RSA_{react}, $SMNA_{react}$, BR_{react}, BRV_{react}	Ratio of average feature values from: 3 1-minute consecutive segments comprising the *Resting* block and 3 1-minute consecutive segments comprising the *Danger* block	
	Recovery	IBI_{recov}, RSA_{recov}, $SMNA_{recov}$, BR_{recov}, BRV_{recov}	Ratio of average feature values from: 3 1-minute consecutive segments comprising the *Danger* block and 2 1-minute consecutive segments comprising the *Recovery* block	

for four features, one with the highest correlation from each group, to visually illustrate their relationships with simulator performance.

B. SIMULATOR PERFORMANCE PREDICTION RESULTS

A LOOCV framework was implemented, within which feature selection and model fitting were performed on the training subset in each fold, and the quality of prediction was evaluated on the left-out participant. The optimization algorithm aimed to identify the best-performing combination of the four predictors by minimizing the RMSE of the multiple regression model fitted to the training data. One feature was selected from each of four distinct theoretical groups: resting physiological features, startle reflex physiological features, allostatic features, and cognitive task performance scores. Across all 15 LOOCV folds, only 7 out of all possible features were selected. Within the startle reflex and allostatic groups, SH_{EMG} and IBI_{recov} were consistently chosen as the optimal features in all 15 folds. In the resting feature group, RSA_4 was selected in 10 out of 15 folds, while BR was chosen in 3 folds and RSA_2 in 2 folds. Within the cognitive task group, dStroopTones was selected in 13 folds, and PVSAT

TABLE 3. An overview of cognitive task scores obtained during Phase 1, and simulator performance metric obtained during Phase 2 on a flight simulator use case.

PHASE 1						PHASE 2
			Dual Stroop			
MOT	PVSAT	Stroop (StroopColours)	Stroop Accuracy (dStroopColours)	Tone Counting Accuracy (dStroopTones)	N-back	Pitch tracking task with startle events (simulator performance)
Percentage of correctly detected moving dots	Percentage of correct answers	Percentage of correctly named font colours	Percentage of correctly named font colours	Performance score based on the sum of differences between given answers and correct ones	Percentage of correct answers	Overall error between the reference and the aircraft pitch angle

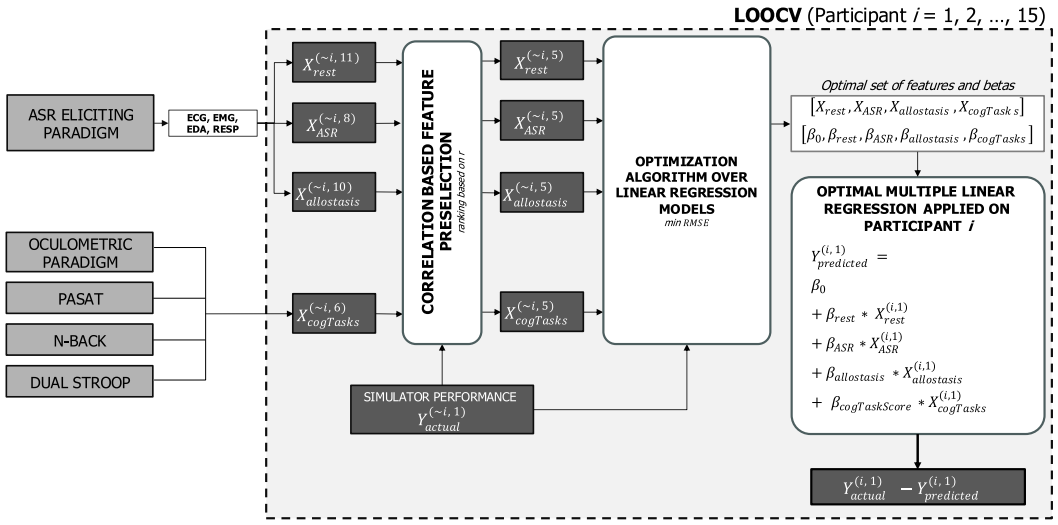


FIGURE 7. Overview of the data analysis approach. Features from four domains (resting physiology, acoustic startle reflex, allostasis, and cognitive performance) were measured in Phase 1. These features were preselected using correlation-based filtering within a leave-one-out cross-validation (LOOCV) framework. In each LOOCV iteration, one participant (i) was held out for testing, while the remaining participants (~ i) were used for model training. An optimization algorithm then selected the best combination of one feature per domain to minimize the root mean square error (RMSE) in a multiple regression model predicting simulator performance from Phase 2. Model performance was evaluated using mean absolute error (MAE) between predicted and actual simulator scores.

was selected in 2 folds. In each LOOCV iteration, a 4-feature multiple regression model that obtained the minimum RMSE when fitted to the training data was evaluated on the left-out participant. Models achieved a normalized MAE of 0.6143, and a MAE of 2.5457° in comparison with a dummy median regressor MAE of 3.4881°. The Pearson correlation between the actual and predicted simulator performance across all LOOCV iterations was $r = 0.7039$ ($p = 0.0034$), indicating a strong and statistically significant predictive relationship. Fig. 9 shows the plot of the actual versus predicted simulator performance across all left-out participants from the LOOCV procedure, including the 45-degree reference line. This plot illustrates the models' predictive alignment with observed values throughout all folds.

Fig. 10 presents two subplots summarizing the behaviour of selected predictors across the LOOCV process. The left subplot shows a bar chart indicating the frequency with which each feature was selected as part of the optimal predictor set across 15 LOOCV iterations. The right subplot displays a boxplot of the beta estimates for each predictor across all folds, providing insight into the variability and magnitude of each predictor's contributions to the models.

V. DISCUSSION

The optimization algorithm selected a set of four stress resilience features and cognitive task scores, one from each proposed group, as the optimal set of predictors in each LOOCV fold. Across the 15 LOOCV folds, the most frequently selected features were: startle reflex habituation (SH_{EMG}) from the startle reflex features and recovery of heart rate after threat (IBI_{recov}) from the allostatic features, both of which were chosen in all 15 folds; resting respiratory sinus arrhythmia (RSA_4) from the resting features, selected in 10 folds; and a multitasking-related score (dStroopTones) from the cognitive performance task scores, selected in 13 folds. Three remaining selected features—BR, RSA_2 and PVSAT—were selected far less frequently (3, 2, and 2 times, respectively), indicating their more limited influence in the predictive model. Therefore, the following discussion will focus only the four most frequently selected features.

TABLE 4. Results of the correlation analysis between selected stress resilience features, cognitive task scores, and simulator performance. The table also shows the original (pre-normalized) value ranges of each feature or score. An upward arrow (↑) indicates that higher values reflect a stronger presence of the measured phenomenon. Conversely, a downward arrow (↓) indicates that lower values reflect a stronger presence of the measured phenomenon.

Group	Feature/Score	Original range	Pearson		Spearman	
			r	p-value	r	p-value
Resting Features	BR ↑	[0.1708, 0.3616]	0.2388	0.3914	0.0786	0.7808
	RSA_4 ↑	[-0.1165, 0.6597]	-0.1932	0.4903	-0.1143	0.6851
	BA ↑	[0.2039, 3.7633]	0.2478	0.3733	0.4107	0.1297
Startle Features	FPS_{EMG} ↑	[0.5495, 5.3607]	-0.3684	0.1766	-0.2143	0.4431
	SH_{EMG} ↓	[-1.465, 1.7622]	0.4918	0.0626	0.4071	0.1320
	SR_{EMG} ↑	[2.5796, 71.5186]	0.1937	0.4890	0.3214	0.2427
Allostatic Features	IBI_{recov} ↑	[-0.1731, 0.0655]	0.2927	0.2897	0.4286	0.1126
	RSA_{react} ↑	[-0.2178, 0.2859]	0.3140	0.2545	0.1679	0.5493
	$SMNA_{react}$ ↑	[0.9976, 17.413]	0.2053	0.4629	0.2857	0.3011
Cognitive Task Scores	PVSAT↑	[75.64, 93.59]	-0.5069	0.0538	-0.5009	0.0572
	StroopColours↑	[0.8222, 1]	-0.3320	0.1600	-0.3592	0.1885
	dStroopTones↑	[85, 99]	-0.5666	0.0277	-0.5085	0.0529

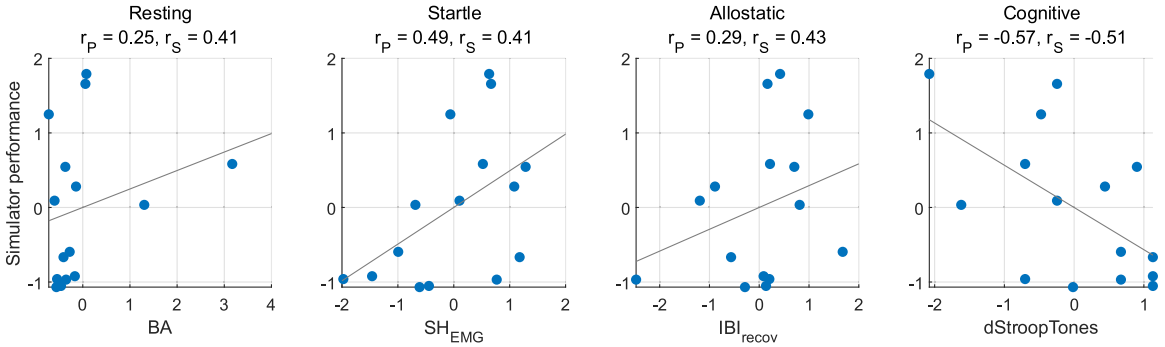


FIGURE 8. Scatter plots of top-performing features from each domain group in Phase 1 (Resting, Startle, Allostatic, and Cognitive), plotted against simulator performance scores from Phase 2. Each plot displays standardized feature values versus standardized pitch-tracking error, along with Pearson (r_P) and Spearman (r_S) correlation coefficients and corresponding least-squares fit lines.

Models' performance was evaluated across the 15 LOOCV folds, with metrics including a Mean Absolute Error (MAE) of 2.5457°, normalized MAE of 0.6143, Pearson's correlation between actual and predicted simulator performance (r) of 0.7039, and a significant p-value of 0.0034. These results indicate that the selected 4-feature sets substantially contributed to the MRA-based prediction of pilot simulator performance under startle conditions.

Beta estimates obtained across LOOCV folds further explained the relative contribution of each predictor. SH_{EMG} exhibited the largest absolute beta values, highlighting it as the most influential feature in predicting simulator performance, followed by IBI_{recov} and dStroopTones. RSA_4, while having the smallest absolute beta estimates, was still frequently selected and emerged as a valuable predictor in combination with other features, despite its low correlation with the pitch-tracking error. This underscores the utility of multiple regression analysis in uncovering unique contributions of individual predictors that may not be revealed by bivariate correlational analyses [89], [90], [91].

Most of the optimally selected features followed trends that were in line with theoretical expectations. A positive trend was observed for SH_{EMG}, indicating that a smaller startle habituation was associated with larger pitch-tracking error throughout the flight simulation with startle events. A negative trend was observed for the features dStroopAcc and RSA_4, suggesting that lower multitasking-related accuracy and lower RSA, which reflects reduced autonomic flexibility and stress regulation, were associated with larger pitch-tracking error. Notably, predictor selected from the cognitive task scores group (dStroopTones) is related to the multitasking Dual Stroop task, which demands greater resource allocation and adaptability than other context-free stimulation paradigms in this paper; both of these capabilities are crucial for pilots managing startle/unexpected events [92]. Multitasking ability has also been previously shown to predict flying performance [80]. However, IBIrecov showed a positive trend that does not fully align with theoretical expectations, because the trend suggests that a faster recovery of cardiac IBI, which indicates higher stress resilience, is associated with larger pitch-tracking error. Although counterintuitive from resilience-theoretic standpoint, this finding may reflect the sympathetic predominance observed in fighter pilots' cardiovascular profiles compared to control groups [93], [94], potentially associated with heightened alertness [95]. However, experimental testing of the relationship between cardiac IBI recovery after generic context-free laboratory stressors and real-world task performance has been neglected so far, to our knowledge, and further experiments are needed to explain this knowledge gap more rigorously.

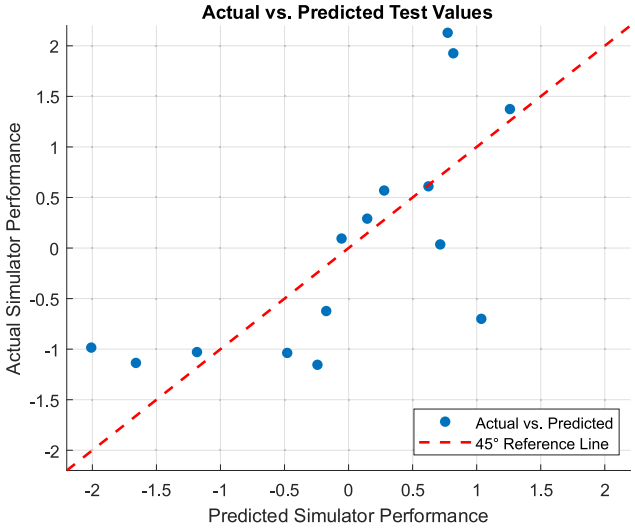


FIGURE 9. Plot of actual vs. predicted simulator performance during LOOCV, with the 45-degree reference line, showing test data points across all folds (MAE = 2.5457°; MAEnorm = 0.6143; r = 0.7039; p = 0.0034).

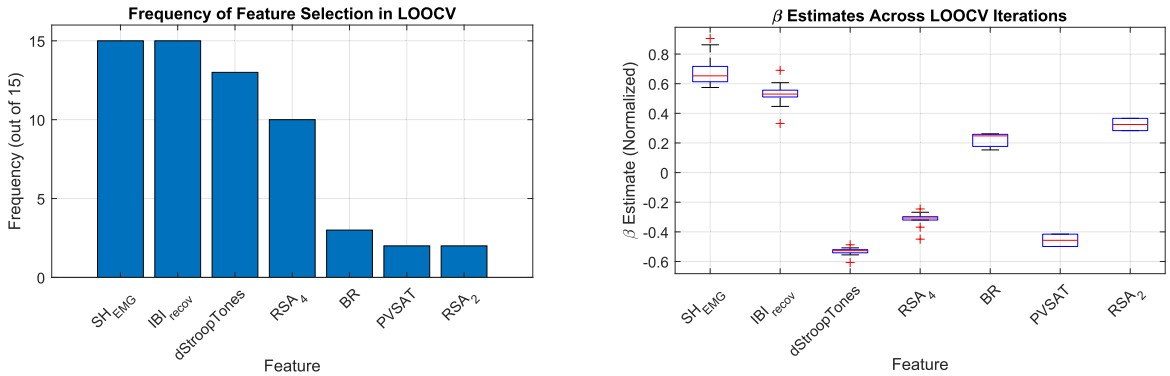


FIGURE 10. Left: Frequency of selection for each predictor in the optimal feature set across 15 LOOCV iterations. Right: Boxplot of standardized beta estimates for each predictor across LOOCV folds, showing the variability in their contribution to the regression.

These findings suggest that pilot simulator performance, particularly under startle conditions, depends on baseline physiology, startle habituation and recovery, as well as multitasking ability and cognitive flexibility. These results are in line with prior stress resilience research linking regulation of the autonomic nervous system and cognitive adaptability to threat responses [26], [96], [97], [98]. The features identified here predicted how well pilots adapted to perform under startle events, which is important given that sudden, intense events (e.g., system failures or alarms) are known to impair decision-making and motor control in critical phases of flight [5], [99], [100]. These findings also align with broader literature emphasizing the role of cognitive flexibility and multitasking in complex operational settings like aviation [101], [102], [103].

While the current study focuses on features derived from out-of-context paradigms and cognitive tasks to predict pilot performance under startle conditions, future research will expand this approach by incorporating in-flight physiological data for real-time pilot performance monitoring and prediction. This integration aims to improve the validity and predictive power of the models by capturing pilots' physiological responses in operational environments. Additionally, other advanced modelling approaches, including more comprehensive machine learning techniques and larger datasets involving a wider range of cognitive, emotional, and motor tasks, are planned for subsequent analyses.

VI. CONCLUSION

This study illustrates the feasibility of selected neuropsychophysiological stress resilience features and cognitive task scores, obtained from context-free stimulation paradigms, as predictors of pilot simulator performance during startle events. First research question (RQ1) explored how well multimodal neuropsychophysiological stress resilience features related to the acoustic startle reflex, resting autonomic activity, and allostasis, combined with cognitive task scores, predict pilot simulator performance during startle events. The results demonstrated that these features, obtained from generic context-free paradigms, can substantially predict the pilot simulator performance. Multiple regression models using these features/scores achieved significant pre-

dictive validity in the LOOCV procedure (r = 0.7039, p = 0.0034, MAE = 2.5457°). Second research question (RQ2) focused on selection of the most influential predictors of pilot simulator performance under startle conditions, and four features emerged as optimal: resting respiratory sinus arrhythmia, startle reflex habituation, heart rate recovery after threat, and a specific score on multitask dual-Stroop paradigm. These features/scores reflect differences in the regulation of the autonomic system and cognitive control capacities critical in maintaining pilot performance in high-stakes startle situations such as unexpected system failures or alarms. The limitations of this study include the relatively small sample size, which limits the generalizability of the findings and a limited exploration of alternative modelling approaches. These limitations will be addressed in future research, as outlined in the Discussion section.

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REFERENCES

[1] R. I. Thackray, "Performance recovery following startle: A laboratory approach to the study of behavioral response to sudden aircraft emergencies," Federal Aviation Admin., Washington, DC, USA, Tech. Rep. DOT/FAA-AM-88/4, 1988, doi: 10.1037/e586272011-001.

[2] Federal Aviation Administration. (Apr. 2015). *Upset Prevention and Recovery Training - With Changes*. Accessed: May, 16, 2025. [Online]. Available: https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentid/1027328

[3] M. Diarra, M. Marchitto, M.-C. Bressolle, T. Baccino, and V. Drai-Zerbib, "A narrative review of the interconnection between pilot acute stress, startle, and surprise effects in the aviation context: Contribution of physiological measurements," *Frontiers Neuroergonomics*, vol. 4, Feb. 2023, Art. no. 1059476, doi: 10.3389/fnrgo.2023.1059476.

[4] J. Rivera, A. B. Talone, C. T. Boesser, F. Jentsch, and M. Yeh, "Startle and surprise on the flight deck: Similarities, differences, and prevalence," in *Proc. Hum. Factors Ergonom. Soc. Annu. Meeting*, vol. 58, Sep. 2014, pp. 1047–1051, doi: 10.1177/1541931214581219.

[5] L. Kinney and D. O'Hare, "Responding to an unexpected in-flight event: Physiological arousal, information processing, and performance," *Hum. Factors, J. Hum. Factors Ergonom. Soc.*, vol. 62, no. 5, pp. 737–750, Aug. 2020, doi: 10.1177/0018720819854830.

[6] N. Clutton, "The science behind startle/surprise and fatigue resilience," *Air Pilot*, vol. 40, pp. 23–25, Aug. 2020.

[7] J. N. Field, E. J. Boland, J. M. van Rooij, J. F. W. Mohrmann, and J. W. Smeltink, "Study for the European aviation safety agency: Startle effect management," Netherlands Aerospace Centre, Amsterdam, The Netherlands, Tech. Rep. NLR-CR-2018-242, 2018. [Online]. Available: https://www.easa.europa.eu/sites/default/files/dfu/EASA_Research_Startle_Effect_Managements_Final_Report.pdf

[8] S. Putz, A. Mertens, L. Chuang, and V. Nitsch, "Physiological measures of operators' mental state in supervisory process control tasks: A scoping review," *Ergonomics*, vol. 67, no. 6, pp. 801–830, Jun. 2024, doi: 10.1080/00140139.2023.2289858.

[9] V. Ferrari, J. Gagnon, C. Camachon, and M. Kopf, "Psycho-physiological evaluation of the pilot: A study conducted with pilots of the French air force," in *Engineering Psychology and Cognitive Ergonomics* (Lecture Notes in Computer Science), vol. 10906, D. Harris, Ed., Cham, Switzerland: Springer, 2018, pp. 285–295, doi: 10.1007/978-3-319-91122-9_24.

[10] H. Mansikka, K. Virtanen, and D. Harris, "Comparison of NASA-TLX scale, modified Cooper–Harper scale and mean inter-beat interval as measures of pilot mental workload during simulated flight tasks," *Ergonomics*, vol. 62, no. 2, pp. 246–254, Feb. 2019, doi: 10.1080/00140139.2018.1471159.

[11] P. A. Hebbar, K. Bhattacharya, G. Prabhakar, A. A. Pashilkar, and P. Biswas, "Correlation between physiological and performance-based metrics to estimate pilots," *Frontiers Psychol.*, vol. 12, Apr. 2021, Art. no. 555446, doi: 10.3389/fpsyg.2021.555446.

[12] K. Mohanavelu, S. Poonguzhali, D. Ravi, P. K. Singh, M. Mahajabin, R. K., U. K. Singh, and S. Jayaraman, "Cognitive workload analysis of fighter aircraft pilots in flight simulator environment," *Defence Sci. J.*, vol. 70, no. 2, pp. 131–139, Mar. 2020, doi: 10.14429/dsj.70.14539.

[13] B. Zhou, L. Ding, B. Chen, H. Shi, Y. Ao, R. Xu, and Y. Li, "Physiological characteristics and operational performance of pilots in the high temperature and humidity fighter cockpit environments," *Sensors*, vol. 21, no. 17, p. 5798, Aug. 2021, doi: 10.3390/s21175798.

[14] E. van Weelden, M. Alimardani, T. J. Wiltshire, and M. M. Louwerse, "Aviation and neurophysiology: A systematic review," *Appl. Ergonom.*, vol. 105, Nov. 2022, Art. no. 103838, doi: 10.1016/j.apergo.2022.103838.

[15] M. Šarlija, S. Popovic, M. Jagodic, T. Jovanovic, V. Ivkovic, Q. Zhang, G. Strangman, and K. Cosic, "Prediction of task performance from physiological features of stress resilience," *IEEE J. Biomed. Health Informat.*, vol. 25, no. 6, pp. 2150–2161, Jun. 2021, doi: 10.1109/JBHI.2020.3041315.

[16] M. Causse, Z. Chua, and N. Matton, "Can efficiency and prefrontal activity during a working memory task predict pilots," in *Proc. Int. Conf.*, Dubrovnik, Croatia, 2018, pp. 1–14.

[17] T. Miyatsu, B. M. Smith, A. P. Koutnik, P. Pirolli, and T. J. Broderick, "Resting-state heart rate variability after stressful events as a measure of stress tolerance among elite performers," *Frontiers Physiol.*, vol. 13, Jan. 2023, Art. no. 1070285, doi: 10.3389/fphys.2022.1070285.

[18] D. Mouratille, F. Amadieu, and N. Matton, "A preliminary study on the usefulness of physiological assessment for the ATCO selection," in *Proc. Int. Symp. Aviation Psychol.*, Corvallis, OR, USA, May 2021, pp. 1–16. [Online]. Available: https://hal.archives-ouvertes.fr/hal-03274237

[19] M. Koch, "The neurobiology of startle," *Prog. Neurobiol.*, vol. 59, no. 2, pp. 107–128, Oct. 1999, doi: 10.1016/s0301-0082(98)00098-7.

[20] T. Jovanovic, S. D. Norrholm, N. Q. Blanding, M. Davis, E. Duncan, B. Bradley, and K. J. Ressler, "Impaired fear inhibition is a biomarker of PTSD but not depression," *Depression Anxiety*, vol. 27, no. 3, pp. 244–251, Mar. 2010, doi: 10.1002/da.20663.

[21] S. D. Norrholm, T. Jovanovic, I. W. Olin, L. A. Sands, I. Karapanou, B. Bradley, and K. J. Ressler, "Fear extinction in traumatized civilians with posttraumatic stress disorder: Relation to symptom severity," *Biol. Psychiatry*, vol. 69, no. 6, pp. 556–563, Mar. 2011, doi: 10.1016/j.biopsych.2010.09.013.

[22] N. Pole, T. C. Neylan, C. Otte, C. Henn-Hasse, T. J. Metzler, and C. R. Marmar, "Prospective prediction of posttraumatic stress disorder symptoms using fear potentiated auditory startle responses," *Biol. Psychiatry*, vol. 65, no. 3, pp. 235–240, Feb. 2009, doi: 10.1016/j.biopsych.2008.07.015.

[23] V. Risbrough, D. G. Baker, C. Nievergelt, B. Litz, W. Nash, J. Perez, and M. Geyer, "Prospective assessment of psychophysiological risk factors for PTSD," *Eur. J. Psychotraumatol.*, vol. 3, pp. 1–12, Sep. 2012, doi: 10.3402/ejpt.v3i0.19473.

[24] D. E. Bradford, J. T. Kaye, and J. J. Curtin, "Not just noise: Individual differences in general startle reactivity predict startle response to uncertain and certain threat," *Psychophysiology*, vol. 51, no. 5, pp. 407–411, May 2014, doi: 10.1111/psyp.12193.

[25] E. Poli and A. Angrilli, "Greater general startle reflex is associated with greater anxiety levels: A correlational study on 111 young women," *Frontiers Behav. Neurosci.*, vol. 9, pp. 1–14, Feb. 2015, doi: 10.3389/fnbeh.2015.00010.

[26] F. R. Walker, K. Pfingst, L. Carnevali, A. Sgoifo, and E. Nalivaiko, "In the search for integrative biomarker of resilience to psychological stress," *Neurosci. Biobehavioral Rev.*, vol. 74, pp. 310–320, Mar. 2017, doi: 10.1016/j.neubiorev.2016.05.003.

[27] E. Nalivaiko, A. Thomson, E. Aidman, and R. Walker, "Habituation of acoustic startle: A new biomarker of psychological resilience," *J. Sci. Med. Sport*, vol. 20, p. S52, Nov. 2017, doi: 10.1016/j.jsams.2017.09.163.

[28] K. Cosic, M. Sarlija, V. Ivkovic, Q. Zhang, G. Strangman, and S. Popovic, "Stress resilience assessment based on physiological features in selection of air traffic controllers," *IEEE Access*, vol. 7, pp. 41989–42005, 2019, doi: 10.1109/ACCESS.2019.2907479.

[29] S. J. Russo, J. W. Murrough, M.-H. Han, D. S. Charney, and E. J. Nestler, "Neurobiology of resilience," *Nature Neurosci.*, vol. 15, no. 11, pp. 1475–1484, Nov. 2012, doi: 10.1038/nn.3234.

[30] M. V. Jones, N. Smith, D. Burns, E. Braithwaite, M. Turner, A. McCann, L. Walker, P. Emmerson, L. Webster, and M. Jones, "A systematic review of resilient performance in defence and security settings," *PLoS ONE*, vol. 17, no. 10, Oct. 2022, Art. no. e0273015, doi: 10.1371/journal.pone.0273015.

[31] L. Li, Y. Chen, and R. Xing, "Influence and mechanism of resilience on flight operational performance under different flight situations: A study based on flight simulator experiments," *Saf. Sci.*, vol. 184, Apr. 2025, Art. no. 106763, doi: 10.1016/j.ssci.2024.106763.

[32] C. Caponecchia, W. Y. Zheng, and M. A. Regan, "Selecting trainee pilots: Predictive validity of the WOMBAT situational awareness pilot selection test," *Appl. Ergonom.*, vol. 73, pp. 100–107, Nov. 2018, doi: 10.1016/j.apergo.2018.06.004.

[33] Y. W. Sohn and S. M. Doane, "Memory processes of flight situation awareness: Interactive roles of working memory capacity, long-term working memory, and expertise," *Human Factors, J. Human Factors Ergonom. Soc.*, vol. 46, no. 3, pp. 461–475, Sep. 2004, doi: 10.1518/hfes.46.3.461.50392.

[34] M. Causse, F. Dehais, and J. Pastor, "Executive functions and pilot characteristics predict flight simulator performance in general aviation pilots," *Int. J. Aviation Psychol.*, vol. 21, no. 3, pp. 217–234, Jul. 2011, doi: 10.1080/10508414.2011.582441.

[35] E. Rosa, E. Lyskov, M. Grönkvist, R. Kölegård, N. Dahlström, I. Knez, R. Ljung, and J. Willander, "Cognitive performance, fatigue, emotional, and physiological strains in simulated long-duration flight missions," *Mil. Psychol.*, vol. 34, no. 2, pp. 224–236, Mar. 2022, doi: 10.1080/08995605.2021.1989236.

[36] L. Ke, Z. Zhang, Y. Ma, Y. Xiao, S. Wu, X. Wang, X. Liu, and J. He, "Evaluating flight performance and eye movement patterns using virtual reality flight simulator," *J. Visualized Experiments*, vol. 2023, no. 195, p. 65170, May 2023, doi: 10.3791/65170.

[37] Y. Hilla, M. Stefani, M. Michael Nitzschner, and W. Mack, "Flight simulation task performance predicts military multitasking better than laboratory measures," presented at the *Proc. presented at the 15th Int. Erence Appl. Hum. Factors Ergonom.*, Jun. 2024, doi: 10.54941/ahfe1004881.

[38] P. Strózak, P. Francuz, R. Lewkowicz, P. Augustynowicz, A. Fudali-Czyz, B. Bałaj, and O. Truszczyński, "Selective attention and working memory under spatial disorientation in a flight simulator," *Int. J. Aerosp. Psychol.*, vol. 28, nos. 1–2, pp. 31–45, Apr. 2018, doi: 10.1080/24721840.2018.1486195.

[39] K. ALMamari and A. Traynor, "Multiple test batteries as predictors for pilot performance: A meta-analytic investigation," *Int. J. Selection Assessment*, vol. 27, no. 4, pp. 337–356, Dec. 2019, doi: 10.1111/ijsa.12258.

[40] M. Šarlija, S. Popović, T. Jovanovic, and K. Cosic, "Temporal stability of resting respiratory sinus arrhythmia: Choosing the quantification method," in *Proc. IEEE Int. Workshop Technol. Defense Secur.*, Rome, Italy, Nov. 2023, pp. 61–65, doi: 10.1109/TECHDEFENSE59795.2023.10380811.

[41] S. J. Vine, L. Uiga, A. Lavric, L. J. Moore, K. Tsaneva-Atanasova, and M. R. Wilson, "Individual reactions to stress predict performance during a critical aviation incident," *Anxiety, Stress, Coping*, vol. 28, no. 4, pp. 467–477, Jul. 2015, doi: 10.1080/10615806.2014.986722.

[42] B. J. Lukey and V. Tepe, *Biobehavioral Resilience To Stress*. Boca Raton, FL, USA: CRC Press, 2008.

[43] A. C. Stolk-Vos, M. H. Heres, J. Kesteloo, D. Verburg, F. Hiddema, D. A. Lie, and D. F. de Korne, "Is there a role for the use of aviation assessment instruments in surgical training preparation? A feasibility study," *Postgraduate Med. J.*, vol. 93, no. 1095, pp. 20–24, Jan. 2017, doi: 10.1136/postgradmedj-2016-133984.

[44] A. Landman, P. van Oorschot, M. M. R. van Paassen, E. L. Groen, A. W. Bronkhorst, and M. Mulder, "Training pilots for unexpected events: A simulator study on the advantage of unpredictable and variable scenarios," *Hum. Factors, J. Hum. Factors Ergonom. Soc.*, vol. 60, no. 6, pp. 793–805, Sep. 2018, doi: 10.1177/0018720818779928.

[45] M. B. Russo, A. P. Kendall, D. Johnson, H. C. Sing, D. R. Thorne, S. M. Escolas, S. Santiago, D. Holland, S. W. Hall, and D. P. Redmond, "Visual perception, psychomotor performance, and complex motor performance during an overnight air refueling simulated flight," *Aviat. Space Environ. Med.*, vol. 76, no. 7 Suppl, pp. C92–103, Jul. 2005.

[46] W. K. Kirchner, "Age differences in short-term retention of rapidly changing information," *J. Experim. Psychol.*, vol. 55, no. 4, pp. 352–358, 1958, doi: 10.1037/h0043688.

[47] D. Schoofs, S. Pabst, M. Brand, and O. T. Wolf, "Working memory is differentially affected by stress in men and women," *Behavioural Brain Res.*, vol. 241, pp. 144–153, Mar. 2013, doi: 10.1016/j.bbr.2012.12.004.

[48] I. Kesedzic, M. Sarlija, J. Bozek, S. Popovic, and K. Cosic, "Classification of cognitive load based on neurophysiological features from functional near-infrared spectroscopy and electrocardiography signals on n-Back task," *IEEE Sensors J.*, vol. 21, no. 13, pp. 14131–14140, Jul. 2021, doi: 10.1109/JSEN.2020.3038032.

[49] J. R. Stroop, "Studies of interference in serial verbal reactions," *J. Experim. Psychol.*, vol. 18, no. 6, pp. 643–662, Dec. 1935, doi: 10.1037/h0054651.

[50] E. K. Miller and J. D. Cohen, "An integrative theory of prefrontal cortex function," *Annu. Rev. Neurosci.*, vol. 24, no. 1, pp. 167–202, Mar. 2001, doi: 10.1146/annurev.neuro.24.1.167.

[51] M. J. M. Lamers, A. Roelofs, and I. M. Rabeling-Keus, "Selective attention and response set in the stroop task," *Memory Cognition*, vol. 38, no. 7, pp. 893–904, Oct. 2010, doi: 10.3758/mc.38.7.893.

[52] H. Pashler, "Dual-task interference in simple tasks: Data and theory," *Psychol. Bull.*, vol. 116, no. 2, pp. 220–244, 1994, doi: 10.1037/0033-2909.116.2.220.

[53] M. Tombu and P. Jolicœur, "A central capacity sharing model of dual-task performance," *J. Experim. Psychol., Human Perception Perform.*, vol. 29, no. 1, pp. 3–18, 2003, doi: 10.1037/0096-1523.29.1.3.

[54] D. Simoni, G. Rubbieri, M. Baccini, L. Rinaldi, D. Becheri, T. Forconi, E. Mossello, S. Zanieri, N. Marchionni, and M. D. Bari, "Different motor tasks impact differently on cognitive performance of older persons during dual task tests," *Clin. Biomechanics*, vol. 28, no. 6, pp. 692–696, Jul. 2013, doi: 10.1016/j.clinbiomech.2013.05.011.

[55] C.-T. Lin, S.-A. Chen, T.-T. Chiu, H.-Z. Lin, and L.-W. Ko, "Spatial and temporal EEG dynamics of dual-task driving performance," *J. NeuroEng. Rehabil.*, vol. 8, no. 1, p. 11, Dec. 2011, doi: 10.1186/1743-0003-8-11.

[56] J. Koerts, C. C. Aarnoudse, and J. M. Spikman, "Paced auditory serial addition test (PASAT)," in *A Compendium of Neuropsychological Tests: Fundamentals of Neuropsychological Assessment and Test Reviews for Clinical Practice*. London, U.K.: Oxford Univ. Press, 2012, p. 298.

[57] R. Allen, P. Mcgeorge, D. Pearson, and A. B. Milne, "Attention and expertise in multiple target tracking," *Appl. Cognit. Psychol.*, vol. 18, no. 3, pp. 337–347, Apr. 2004, doi: 10.1002/acp.975.

[58] H. S. Meyerhoff, F. Papenmeier, and M. Huff, "Studying visual attention using the multiple object tracking paradigm: A tutorial review," *Attention, Perception, Psychophys.*, vol. 79, no. 5, pp. 1255–1274, Jul. 2017, doi: 10.3758/s13414-017-1338-1.

[59] K. Cosic, S. Popovic, M. Šarlija, I. Mijic, M. Kokot, I. Kesedžic, G. Strangman, V. Ivkovic, and Q. Zhang, "New tools and methods in selection of air traffic controllers based on multimodal psychophysiological measurements," *IEEE Access*, vol. 7, pp. 174873–174888, 2019, doi: 10.1109/ACCESS.2019.2957357.

[60] K. Ćosić, S. Popović, D. Kukolja, B. Dropuljić, D. Ivanec, and M. Tonković, "Multimodal analysis of startle type responses," *Comput. Methods Programs Biomed.*, vol. 129, pp. 186–202, Jun. 2016, doi: 10.1016/j.cmpb.2016.01.002.

[61] Y. Lu, J. Wu, and S. Fu, "A study of the relationship between novice Pilots' performance and multi-physiology signals," in *Engineering Psychology and Cognitive Ergonomics* (Lecture Notes in Computer Science), vol. 8532. Cham, Switzerland: Springer, 2014, pp. 344–350, doi: 10.1007/978-3-319-07515-0_35.

[62] M.-J. Darvishi-Bayazi, A. Law, S. M. Romero, S. Jennings, I. Rish, and J. Faubert, "Beyond performance: The role of task demand, effort, and individual differences in ab initio pilots," *Sci. Rep.*, vol. 13, no. 1, p. 14035, Aug. 2023, doi: 10.1038/s41598-023-41427-4.

[63] N. B. Sarter, R. J. Mumaw, and C. D. Wickens, "'Pilots' monitoring strategies and performance on automated flight decks: An empirical study combining behavioral and eye-tracking data," in *Human Error Aviation*, 1st ed., R. K. Dismukes, Ed., Evanston, IL, USA: Routledge, 2017, pp. 205–215, doi: 10.4324/9781315092898-12.

[64] S. Pütz, A. Mertens, L. L. Chuang, and V. Nitsch, "Physiological predictors of operator performance: The role of mental effort and its link to task performance," *Hum. Factors, J. Hum. Factors Ergonom. Soc.*, vol. 67, no. 6, pp. 595–615, Jun. 2025, doi: 10.1177/00187208241296830.

[65] I. Alreshidi, D. Bisandu, and I. Moulitsas, "Illuminating the neural landscape of pilot mental states: A convolutional neural network approach with Shapley additive explanations interpretability," *Sensors*, vol. 23, no. 22, p. 9052, Nov. 2023, doi: 10.3390/s23229052.

[66] I. Alreshidi, I. Moulitsas, and K. W. Jenkins, "Advancing aviation safety through machine learning and psychophysiological data: A systematic review," *IEEE Access*, vol. 12, pp. 5132–5150, 2024, doi: 10.1109/ACCESS.2024.3349495.

[67] B. Binias, D. Myszor, and K. A. Cyran, "A machine learning approach to the detection of Pilot's reaction to unexpected events based on EEG signals," *Comput. Intell. Neurosci.*, vol. 2018, pp. 1–9, May 2018, doi: 10.1155/2018/2703513.

[68] Z. Cui, D. Chen, and D. Wu, "EEG-based pilot competency level evaluation in full flight simulator training about abnormal events," in *Proc. IEEE 3rd Int. Conf. Civil Aviation Saf. Inf. Technol. (ICCASIT)*, Oct. 2021, pp. 495–500, doi: 10.1109/ICCASIT53235.2021.9633467.

[69] V. César Cavalcanti Roza and O. Adrian Postolache, "Multimodal approach for emotion recognition based on simulated flight experiments," *Sensors*, vol. 19, no. 24, p. 5516, Dec. 2019, doi: 10.3390/s19245516.

[70] A. R. Harrivel, C. L. Stephens, R. J. Milletich, C. M. Heinich, M. C. Last, N. J. Napoli, N. Abraham, L. J. Prinzel, M. A. Motter, and A. T. Pope, "Prediction of cognitive states during flight simulation using multimodal psychophysiological sensing," in *Proc. AIAA Inf. Syst. AIAA Infotech Aerosp.*, Jan. 2017, pp. 1–16, doi: 10.2514/6.2017-1135.

[71] O. W. Klaproth, C. Vernaleken, L. R. Krol, M. Halbruegge, T. O. Zander, and N. Russwinkel, "Tracing Pilots' situation assessment by neuroadaptive cognitive modeling," *Frontiers Neurosci.*, vol. 14, p. 795, Aug. 2020, doi: 10.3389/fnins.2020.00795.

[72] B. Binias, D. Myszor, H. Palus, and K. A. Cyran, "Prediction of Pilot's reaction time based on EEG signals," *Frontiers Neuroinform.*, vol. 14, p. 6, Feb. 2020, doi: 10.3389/fninf.2020.00006.

[73] H. Mansikka, P. Simola, K. Virtanen, D. Harris, and L. Oksama, "Fighter pilots' heart rate, heart rate variation and performance during instrument approaches," *Ergonomics*, vol. 59, no. 10, pp. 1344–1352, Oct. 2016, doi: 10.1080/00140139.2015.1136699.

[74] G. Borghini, L. Astolfi, G. Vecchiato, D. Mattia, and F. Babiloni, "Measuring neurophysiological signals in aircraft pilots and car drivers for the assessment of mental workload, fatigue and drowsiness," *Neurosci. Biobehavioral Rev.*, vol. 44, pp. 58–75, Jul. 2014, doi: 10.1016/j.neubiorev.2012.10.003.

[75] Z. Wang, L. Zheng, Y. Lu, and S. Fu, "Physiological indices of pilots' abilities under varying task demands," *Aerosp. Med. Human Perform.*, vol. 87, no. 4, pp. 375–381, Apr. 2016, doi: 10.3357/amhp.4386.2016.

[76] X. Hu and G. Lodewijks, "Detecting fatigue in car drivers and aircraft pilots by using non-invasive measures: The value of differentiation of sleepiness and mental fatigue," *J. Saf. Res.*, vol. 72, pp. 173–187, Feb. 2020, doi: 10.1016/j.jsr.2019.12.015.

[77] J. A. Crijnen, "Predicting a pilot,'s cognitive state from physiological measurements," Ph.D thesis, Dept. Cogn. Sci. Artif. Intell., Tilburg Univ., Tilburg, The Netherlands, 2019. [Online]. Available: https://arno.uvt.nl/show.cgi?fid=149399

[78] J. Deniel, M. Dupuy, A. Duchevet, N. Matton, J.-P. Imbert, and M. Causse, "An in-depth examination of mental incapacitation and startle reflex: A flight simulator study," in *Engineering Psychology and Cognitive Ergonomics* (Lecture Notes in Computer Science), D. Harris and W.-C. Li, Eds., Cham, Switzerland: Springer, 2023, pp. 46–59, doi: 10.1007/978-3-031-35389-5_4.

[79] A. R. Hidalgo-Muñoz, D. Mouratille, R. El-Yagoubi, Y. Rouillard, N. Matton, and M. Causse, "Conscientiousness in pilots correlates with electrodermal stability: Study on simulated flights under social stress," *Safety*, vol. 7, no. 2, p. 49, Jun. 2021, doi: 10.3390/safety7020049.

[80] L. G. Barron and M. R. Rose, "Multitasking as a predictor of pilot performance: Validity beyond serial single-task assessments," *Mil. Psychol.*, vol. 29, no. 4, pp. 316–326, Apr. 2017, doi: 10.1037/mil0000168.

[81] M. Martinussen, O. C. Lang-Ree, H. Mjøen, B. Svendsen, and A. Barone, "Predicting commercial pilot training performance: A validation study," *Aviation Psychol. Appl. Hum. Factors*, vol. 12, no. 2, pp. 109–118, Sep. 2022, doi: 10.1027/2192-0923/a000232.

[82] P. R. Jenkins, W. N. Caballero, and R. R. Hill, "Predicting success in United States air force pilot training using machine learning techniques," *Socio-Econ. Planning Sci.*, vol. 79, Feb. 2022, Art. no. 101121, doi: 10.1016/j.seps.2021.101121.

[83] L. G. Barron, T. R. Carretta, and M. V. A. Bonto-Kane, "Relations of personality traits to military aviator performance: It depends on the criterion," *Aviation Psychol. Appl. Human Factors*, vol. 6, no. 2, pp. 57–67, Sep. 2016, doi: 10.1027/2192-0923/a000100.

[84] D. M. Dehn and M. Damitz, "Do personality traits predict training performance?: The use of a personality questionnaire for air traffic controller selection," *Aviation Psychol. Appl. Human Factors*, vol. 12, no. 2, pp. 72–83, Sep. 2022, doi: 10.1027/2192-0923/a000222.

[85] M. Gambiraža, I. Kesedžic, M. Šarlija, S. Popovic, and K. Cosic, "Classification of cognitive load based on oculometric features," in *Proc. 44th Int. Conv. Inf., Commun. Electron. Technol.*, Sep. 2021, pp. 377–382, doi: 10.23919/mipro52101.2021.9597067.

[86] M. Kurtak, M. Gambiraza, M. Vrdoljak, K. K. Nikolić, M. Andrić, M. Valpovac, V. Kopilas, S. Popović, and K. Ćosić, "Startle effect analysis based on multimodal physiological features in a fighter aircraft simulator," presented at Int. Conf. Cogn. Aircr. Syst., Toulouse, France, 2024.

[87] J. I. Daoud, "Multicollinearity and regression analysis," *J. Phys., Conf. Ser.*, vol. 949, Dec. 2017, Art. no. 012009, doi: 10.1088/1742-6596/949/1/012009.

[88] D. L. Banks and S. E. Fienberg, "Statistics, multivariate," in *Encyclopedia of Physical Science and Technology*. Amsterdam, The Netherlands: Elsevier, 2003, pp. 851–889, doi: 10.1016/B0-12-227410-5/00731-6.

[89] Cohen, *Applied Multiple Regression/Correlation Analysis for the Behavioral Sciences*, 3rd ed., Evanston, IL, USA: Routledge, 2013, doi: 10.4324/9780203774441.

[90] R. L. Piedmont, "Multiple regression," in *Encyclopedia of Quality of Life and Well-Being Research*, A. C. Michalos, Ed., Dordrecht, The Netherlands: Springer, 2014, p. 4210, doi: 10.1007/978-94-007-0753-5_1884.

[91] B. Thompson, *Foundations of Behavioral Statistics: An Insight-based Approach*. New York, NY, USA: Guilford Press, 2006.

[92] P. A. Hancock, J. Cruit, J. A. Kochan, M. Gruber, Y. Diaz, A. Kaplan, and S. Pruchnicki, "Relevant research assessment concerning pilot response to unexpected events. Task 2: Relevant research assessment," Federal Aviation Admin., Univ. Central Florida, Washington, DC, USA, Tech. Rep. dot_67402_DS1, Feb. 2022, doi: 10.21949/1528625.

[93] G. B. de Sá, G. D. Rodrigues, and P. P. D. S. Soares, "Cardiovascular autonomic regulation in fighter pilots: Lessons from active standing tests," *Clin. Physiol. Funct. Imag.*, vol. 43, no. 1, pp. 20–27, Jan. 2023, doi: 10.1111/cpf.12788.

[94] A. K. P. Babu, "A comparison of heart rate variability between fighter and non-fighter military aircrew in relation to job stress," Ph.D thesis, Inst. Aerosp. Med. Bangalore, Rajiv Gandhi Univ. Health Sci. (RGUHS), Bengaluru, India. [Online]. Available: https://www.proquest.com/openview/a07900ba3bcc1142cb28af91210920a5/1?cbl=2026366&diss=y&pq-origsite=gscholar

[95] C. Dussault, L. Lely, C. Langrume, F. Sauvet, and J.-C. Jouanin, "Heart rate and autonomic balance during stand tests before and after fighter combat missions," *Aviation, Space, Environ. Med.*, vol. 80, no. 9, pp. 796–802, Sep. 2009, doi: 10.3357/asem.2494.2009.

[96] M. H. Buenrostro-Jáuregui, S. Muñóz-Sánchez, J. Rojas-Hernández, A. I. Alonso-Orozco, G. Vega-Flores, A. Tapia-de-Jesús, and P. Leal-Galicia, "A comprehensive overview of stress, resilience, and neuroplasticity mechanisms," *Int. J. Mol. Sci.*, vol. 26, no. 7, p. 3028, Mar. 2025, doi: 10.3390/ijms26073028.

[97] L. K. Hildebrandt, C. McCall, H. G. Engen, and T. Singer, "Cognitive flexibility, heart rate variability, and resilience predict fine-grained regulation of arousal during prolonged threat," *Psychophysiology*, vol. 53, no. 6, pp. 880–890, Jun. 2016, doi: 10.1111/psyp.12632.

[98] G. Perna, A. Riva, A. Defillo, E. Sangiorgio, M. Nobile, and D. Caldirola, "Heart rate variability: Can it serve as a marker of mental health resilience?" *J. Affect. Disorders*, vol. 263, pp. 754–761, Feb. 2020, doi: 10.1016/j.jad.2019.10.017.

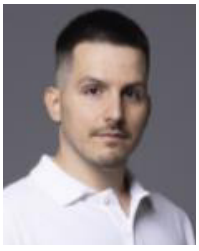
[99] W. L. Martin, P. S. Murray, P. R. Bates, and P. S. Y. Lee, "Fear-potentiated startle: A review from an aviation perspective," *Int. J. Aviation Psychol.*, vol. 25, no. 2, pp. 97–107, Apr. 2015, doi: 10.1080/10508414.2015.1128293.

[100] A. Landman, E. L. Groen, M. M. R. van Paassen, A. W. Bronkhorst, and M. Mulder, "Dealing with unexpected events on the flight deck: A conceptual model of startle and surprise," *Human Factors, J. Human Factors Ergonom. Soc.*, vol. 59, no. 8, pp. 1161–1172, Dec. 2017, doi: 10.1177/0018720817723428.

[101] A. Flood and R. J. Keegan, "Cognitive resilience to psychological stress in military personnel," *Frontiers Psychol.*, vol. 13, Mar. 2022, Art. no. 809003, doi: 10.3389/fpsyg.2022.809003.

[102] Z. Yanzeng, Z. Keyong, C. Hongmin, L. Ziyu, L. Pengyu, and W. Lijing, "The mechanisms linking perceived stress to pilots' safety attitudes: A chain mediation effect of job burnout and cognitive flexibility," *Frontiers Public Health*, vol. 12, Jun. 2024, Art. no. 1342221, doi: 10.3389/fpubh.2024.1342221.

[103] L. Rademacher, D. Kraft, C. Eckart, and C. J. Fiebach, "Individual differences in resilience to stress are associated with affective flexibility," *Psychol. Res.*, vol. 87, no. 6, pp. 1862–1879, Sep. 2023, doi: 10.1007/s00426-022-01779-4.



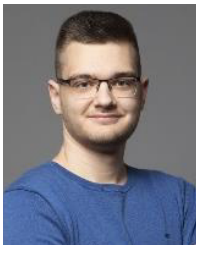
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